

Passive Radar Systems

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ABSTRACT

An essential idea behind passive radar systems is that radar systems of such type do not transmit signal. Instead they rely on transmitter of opportunity like TV, radio, and cellular transmissions. This special operating mechanism have advantages and disadvantages.

This paper discusses passive radar properties, its structure, and the received signal components.

Keywords- Passive radar, Active radar, Spurious Free Dynamic Range (SFDR), Direct Path Interference (DPI), Bistatic/ Multistatic.

INTRODUCTION

Radar: is an EM system for detecting and locating objects. In general radars can be categorized in to Active and Passive.

Active radars operate as follow:

- Transmitting EM signal (controllable).
- Receiving echoes reflected from the target.

- process the echo signal and extract information related to location and velocity of the target.

Passive radars do not transmit signals.

Instead, passive radars relay on signals generated by another source (transmission of opportunity) like TV, Radio broadcasting, and cellular transmission.

Passive Radar Properties

Since passive radars do not transmit signal, there are many advantages and disadvantages occur:

Advantages:

- Hard to be detected.
- No need for transmitting hardware.

- Less cost for maintaining.
- Less power consumption.
- No need for locating a special spectrum for operating.

Disadvantages:

- Complex processing required.
- Power hungry processor.
- Multi-channel system required.

Moreover, there are other factors that decrease passive radar capability of detecting targets:

- Non-coherency between received channels.
- Decreased SFDR of the receiver.

Basic Passive Radar Architecture

- **Two antennas and received channels.**
 - Reference channel, to receive the signal transmitted by the source.
 - Surveillance channel, to receive the signal reflected from the target.

The receiver design has to ensure that the received signal is not falling below the SFDR of the radar.

○ **Processing unit.**

The processing chain is to be detailed later.

Passive Radar In-context

For regular radar systems (Active) they considered to be monostatic (transmitter and receiver are at the same position) and the most well-known types of active radar are: Pulse wave, and continuous wave (CW).

For passive radar they considered to be bistatic (transmitter and receiver are at different positions) CW radar.

Received Signal Components and antenna configuration

The signal received by a passive radars system is mainly constructed of three components:

- **The signal received directly from the transmitter** known as Direct Path Interference (DPI). It has no delay or doppler shift and is the strongest signal.
- **Clutter signal reflected from large stationary obstacles** (buildings, mountains), it is the second strongest signal and low doppler shift.
- **Signal reflected from moving target** which has low amplitude and significant doppler shift due to the high velocity of the target.

Passive Radar Processing

The goal is to detect the signal reflected from a target (with the lowest amplitude) while DPI and the cluttering signal exist. Therefore, the following processing chain executed:

- **Matched Filtering.**

The filter coefficient for matched filter is given by:

$$H(t) = r'^*(-t)$$

r' : is delayed modulated version of r .

The resolution of a passive radar system (ΔR) can be calculated by the formula:

$$\Delta R = \frac{C}{2B}$$

C: the light velocity.

B: the bandwidth.

- **DPI Cancellation.**

The sidelobe of strong signal components mask the desirable reflection. In order to remove the sidelobes all large unwanted components have to be subtracted from the surveillance channel:

$$S' = S - Ax'$$

S' : surveillance channel.

A: matrix of signal components that are removed.

X': vector of weight estimating the contribution of each signal component on **A**.

- **CFAR detection.**

CFAR is well known automatic thresholding scheme. The technique is based on likelihood ratio hypothesis.

REFERENCES

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